

Osterquill Toxicology Inc

CVL kinase series: cardiac ion-channel and CYP interaction triage

Final Report

Study OT-20-S0178 (LIMS OQ/2020/002872)

Client: Corvenlea (US)

Study lead: Dr N. Takamori, Head of Toxicology

Report date: October 20, 2020

1 Summary

40 compounds from the Corvenlea aminopyrimidine series 01 were tested for Corvenlea (US) in the hERG (IKr) safety pharmacology, CYP safety interaction panel. Testing started on October 6, 2020 and results were reported on October 20, 2020. All accepted runs met control validity criteria. No compound had a hERG IC50 below 1 μ M. Most compounds (25 of 40) inhibited hERG with IC50 below 10 μ M, so margins will depend on free therapeutic exposure. Inhibition above 50% at 10 μ M was most frequent for CYP2C9.

Table 1 Study summary

Metric	Value
Project	OT-20-S0178
Client	Corvenlea (US)
Compounds tested	40
Series / chemotype	Corvenlea aminopyrimidine series 01 (Aminopyrimidine)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	14
Price	USD 22,000
hERG IC50 <1 μ M	0
hERG IC50 1-10 μ M	25
hERG IC50 $\geq$ 10 μ M	15
hERG median IC50 (μ M)	8.8
E-4031 control IC50 (nM)	13
CYP1A2 >50% inhibition @10 μ M	7
CYP2C9 >50% inhibition @10 μ M	13
CYP2C19 >50% inhibition @10 μ M	9
CYP2D6 >50% inhibition @10 μ M	3
CYP3A4 >50% inhibition @10 μ M	11
Compounds with any CYP >50%	18
Ketoconazole (3A4) IC50 (μ M)	0.030

Compliance statement

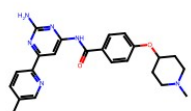
This exploratory in vitro investigation is not conducted under GLP. Procedures follow Osterquill standard operating procedures, and Quality Assurance audits the data for integrity only; no GLP compliance statement is made.

2 Objective

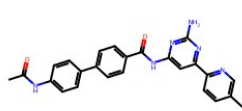
To rank the test article cohort for hERG (IKr) safety pharmacology, CYP safety interaction panel liability so that Corvenlea (US) could choose analogues for progression.

3 Test items

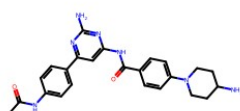
40 compounds were received as 10 mM DMSO solutions in one test article cohort on October 6, 2020 and logged against OQ/2020/002872. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



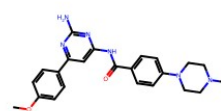
CVL-08810



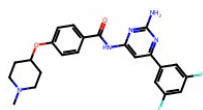
CVL-08818



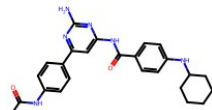
CVL-08824



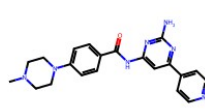
CVL-08828



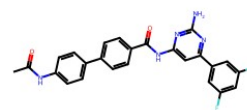
CVL-08834



CVL-08836



CVL-08840



CVL-08846

4 Methods

hERG (IKr) safety pharmacology

HEK293 cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarized to +40 mV for 2 s and repolarized to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Control validity: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- E-4031 IC₅₀ within 5-50 nM; runs outside this range are rejected and repeated
- IC₅₀ fitted per cell (four-parameter logistic); the compound IC₅₀ is the geometric mean of finite cell IC₅₀s when these are the majority of accepted cells, otherwise reported as >30 μ M
- Where precipitation is seen at 30 μ M, that concentration is excluded and the highest evaluable concentration is reported

CYP safety interaction panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m, in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO) incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC₅₀ falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10 μ M are flagged, and small negative values indicate no inhibition.

5 Results

hERG results

All 40 compounds were tested. No compound had a hERG IC₅₀ below 1 μ M. Most compounds (25 of 40) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure. The median IC₅₀ was 8.8 μ M (0 below 1 μ M, 25 between 1 and 10 μ M, 15 at or above 10 μ M, of which 4 above the highest evaluable concentration). E-4031 gave an IC₅₀ of 13 nM in the accepted runs.

Compounds with QC notes: CVL-08882: Precipitation at 30 μ M; highest evaluable concentration 10 μ M (solubility-limited); CVL-08912: Precipitation at 30 μ M; 30 μ M point excluded from fits. Values for these compounds should be read with the note.

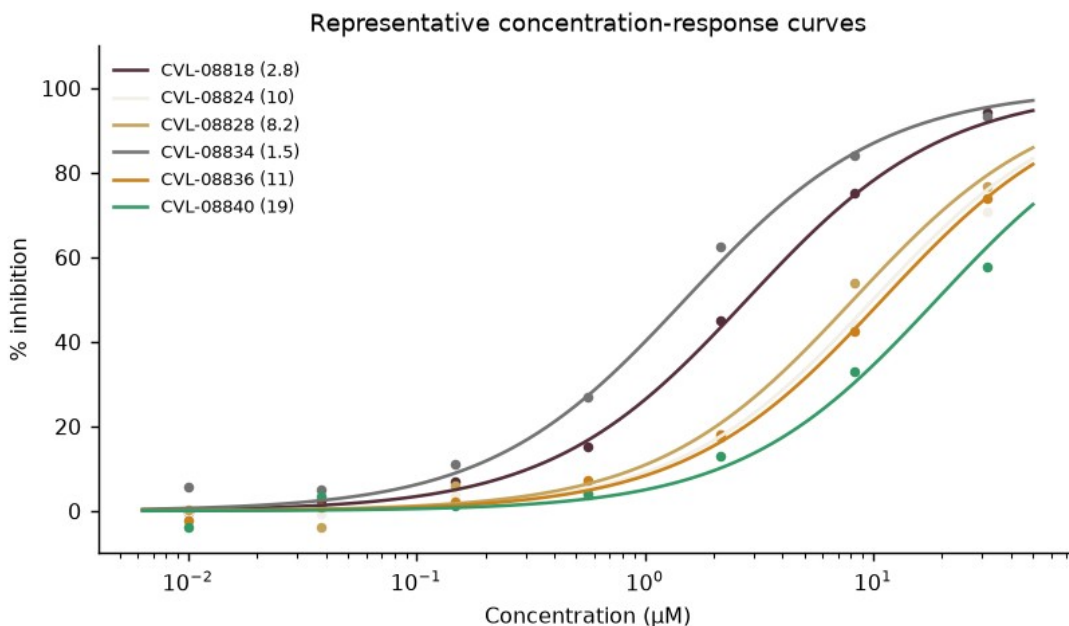
Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
CVL-08810	>30	>30		>30	28	2 (0)	
CVL-08818	2.8	3.5	3.6	3.3	72	3 (3)	
CVL-08824	10	10	6.4	8.7	60	3 (3)	

CVL-08828	8.2	12		9.9	51	2 (2)	
CVL-08834	1.5	2.0		1.7	87	2 (2)	
CVL-08836	11	13		12	49	2 (2)	
CVL-08840	19	11	11	13	43	3 (3)	
CVL-08846	4.1	3.1		3.6	77	2 (2)	
CVL-08848	9.9	8.4		9.1	55	2 (2)	
CVL-08854	27	19		23	28	2 (2)	
CVL-08858	7.4	5.3		6.3	69	2 (2)	
CVL-08860	25	>30		>30	29	2 (1)	
CVL-08866	4.6	8.9		6.4	64	2 (2)	
CVL-08870	27	>30		>30	16	2 (1)	
CVL-08878	2.4	1.6	2.1	2.0	88	3 (3)	
CVL-08882	>30	>30		>10	26	2 (0)	Precipitation at 30 μ M; highest evaluable concentration 10 μ M (solubility-limited)
CVL-08884	23	29	20	24	28	3 (3)	
CVL-08888	4.3	2.0		3.0	78	2 (2)	
CVL-08890	5.0	6.9		5.8	65	2 (2)	
CVL-08894	3.7	2.6		3.1	69	2 (2)	
CVL-08896	9.3	18		13	43	2 (2)	
CVL-08900	23	>30	26	24	22	3 (2)	
CVL-08906	7.3	4.6	12	7.3	54	3 (3)	
CVL-08912	3.4	1.9		2.5	71	2 (2)	Precipitation at 30 μ M; 30 μ M point excluded from fits
CVL-08920	9.7	8.6		9.1	46	2 (2)	
CVL-08924	4.6	2.9		3.6	75	2 (2)	
CVL-08926	9.5	8.3		8.9	56	2 (2)	
CVL-08932	9.3	21		14	45	2 (2)	
CVL-08936	6.3	6.8		6.5	59	2 (2)	
CVL-08938	15	17	12	15	37	3 (3)	
CVL-08944	1.9	1.9		1.9	80	2 (2)	
CVL-08948	5.9	8.7		7.1	64	2 (2)	
CVL-08954	13	13		13	45	2 (2)	
CVL-08960	7.3	4.8		5.9	58	2 (2)	
CVL-08966	10	10		10	47	2 (2)	
CVL-08968	5.4	5.8		5.6	63	2 (2)	
CVL-08978	1.4	2.1		1.7	84	2 (2)	
CVL-08980	7.9	11		9.3	58	2 (2)	
CVL-08984	4.9	6.5		5.7	60	2 (2)	
CVL-08986	19	14		16	36	2 (2)	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 13 nM (acceptance 5-50 nM)

Compound IC₅₀ = geometric mean of finite cell IC₅₀s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)



CYP inhibition results

Inhibition above 50% at 10 µM was most frequent for CYP2C9. In total 18 compounds inhibited at least one isoform by more than 50% at 10 µM (CYP3A4: 11; CYP2D6: 3; CYP2C19: 9; CYP2C9: 13; CYP1A2: 7). All control IC₅₀s were within 0.5-2 times the historical mean (ketoconazole 0.030 µM against a historical mean of 0.030 µM).

Table 3 Percent inhibition at 10 µM

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
CVL-08810	3	24	22	4	53
CVL-08818	20	70	7	4	27
CVL-08824	4	-2	0	8	-6
CVL-08828	6	-6	27	-1	20
CVL-08834	6	75	83	60	40
CVL-08836	7	39	41	4	46
CVL-08840	-11	0	-6	1	5
CVL-08846	33	69	42	77	17
CVL-08848	-2	-2	1	-6	6
CVL-08854	-2	10	-0	-8	15
CVL-08858	17	15	4	-4	-4
CVL-08860	5	3	1	7	2
CVL-08866	60	50	29	6	10
CVL-08870	9	-7	5	-12	5
CVL-08878	10	3	10	5	11
CVL-08882	13	5	62	-8	28
CVL-08884	0	5	-0	-1	12
CVL-08888	5	52	19	-4	25
CVL-08890	30	38	44	22	36
CVL-08894	19	67	20	3	73
CVL-08896	5	-1	1	-4	-4
CVL-08900	8	-6	5	-4	2
CVL-08906	40	11	7	13	45
CVL-08912	7	7	44	0	7

CVL-08920	-5	-6	-4	-4	0
CVL-08924	7	27	15	-2	12
CVL-08926	98	85	80	100	93
CVL-08932	-3	-7	-10	4	-9
CVL-08936	-1	6	2	-5	6
CVL-08938	-0	3	6	6	2
CVL-08944	23	77	78	10	91
CVL-08948	52	50	68	28	91
CVL-08954	11	11	26	0	68
CVL-08960	45	62	51	18	100
CVL-08966	87	77	96	27	90
CVL-08968	14	81	64	9	26
CVL-08978	56	58	62	26	68
CVL-08980	-2	-1	2	3	-2
CVL-08984	58	24	37	28	84
CVL-08986	88	53	25	9	87

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.35; ticlopidine (CYP2C19) 1.1; quinidine (CYP2D6) 0.049; ketoconazole (CYP3A4) 0.030

Table 4 Positive control IC₅₀s

Control	Isoform	IC ₅₀ (μ M)	Historical mean (μ M)
Furafylline	CYP1A2	1.9	1.8
Sulfaphenazole	CYP2C9	0.35	0.35
Ticlopidine	CYP2C19	1.1	1.2
Quinidine	CYP2D6	0.049	0.060
Ketoconazole	CYP3A4	0.030	0.030

6 Deviations

No deviations from the study plan. Precipitation at the top concentration is noted against the affected compounds in Table 2.

7 Discussion

No compound had a hERG IC₅₀ below 1 μ M. Most compounds (25 of 40) inhibited hERG with IC₅₀ below 10 μ M, so margins will depend on free therapeutic exposure. Inhibition above 50% at 10 μ M was most frequent for CYP2C9. Results are exploratory and support compound selection; they are not a substitute for a definitive GLP study.

8 Conclusion

Recommended next steps: prioritize compounds with hERG IC₅₀ at or above 10 μ M; for any compound below 10 μ M, compare the IC₅₀ with projected free plasma exposure before progression; repeat the flagged hERG measurements if a selection decision depends on them; run concentration responses for compounds flagged on CYP2C9 and any other isoform above 50%.

9 Records

Raw data, observation sheets and instrument files are retained for 10 years under OQ/2020/002872.

Name	Role	Signature	Date
Naomi Takamori	Head of Toxicology	<u>N Takamori</u>	October 20, 2020
Lucia Moretti	GLP Quality Assurance and Archive Lead	<u>L Moretti</u>	October 19, 2020